

We built a working lab instrument without CAD software—generative AI designed every part

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Open repository
& full design history

The challenge

Self-driving laboratories let robots run experiments and let software decide what to try next. They handle **liquids** well, but most still weigh out **powders by hand**. Machines that dose powders automatically exist, but they cost **tens to hundreds of thousands of dollars**, putting them out of reach for most academic labs.

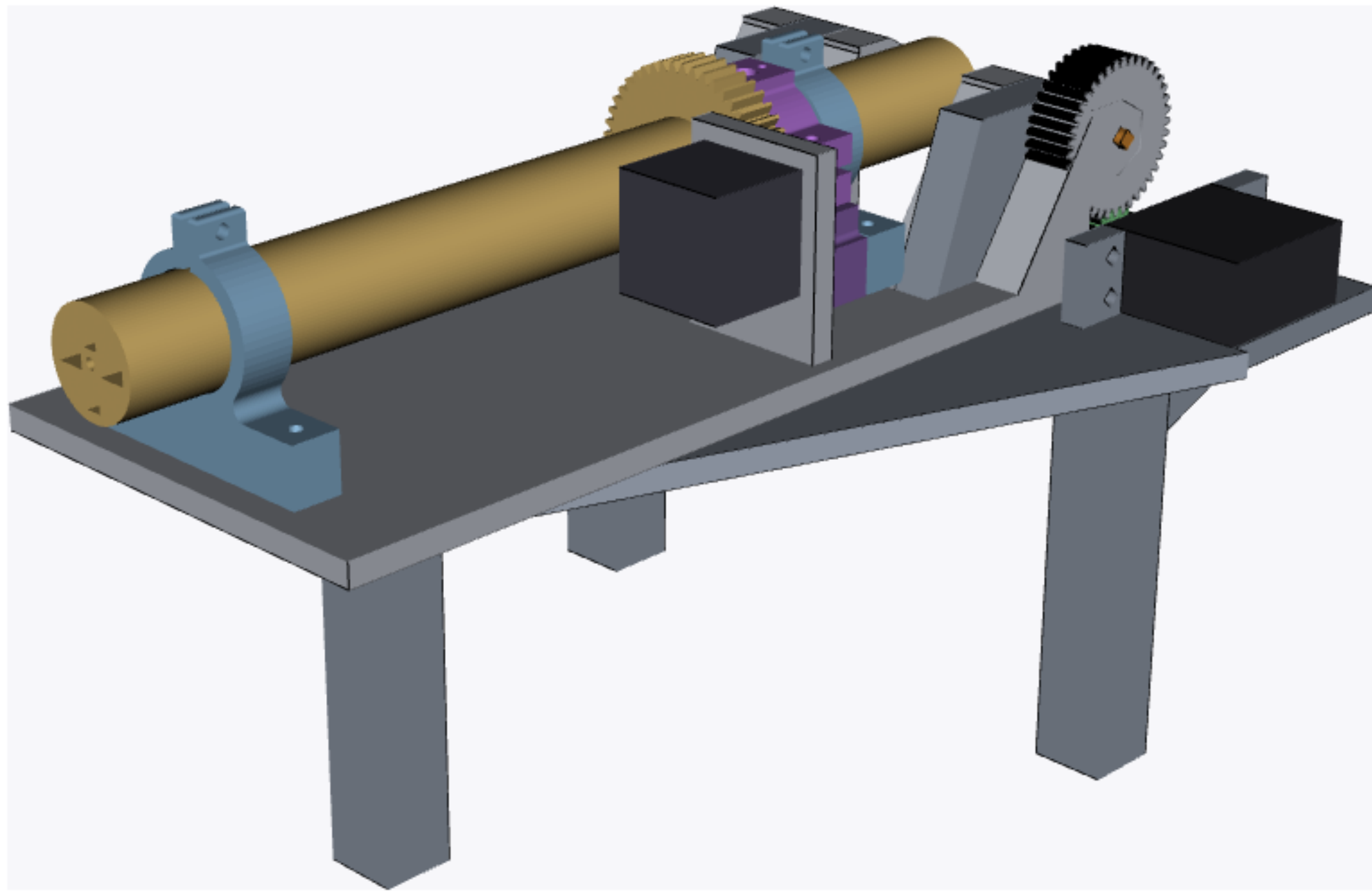
Our question

Generative AI can already write code and produce 3D shapes. We asked a practical question:
Can today’s AI design a real, working physical instrument from start to finish—if a small human team sets the goals and checks the work?
To make the test honest, we set one strict rule.

The rule

No conventional CAD software (SolidWorks, Fusion 360) was used at any point.

Generative AI designed every part of a real, working laboratory instrument—and the complete, honest design history, including its failures, is public.



The single-channel powder doser. Humans chose the design and reviewed, printed, and tested every part; AI tools created all of the 3D models.

97

logged design iterations,
including every failure

0

uses of conventional CAD
software

<\$200

in parts; a \$6 microcontroller
runs it

What we did

We worked in **two AI ways**:

- 1. Programmatic CAD.** AI “coding agents” wrote each part as computer code; the code became a 3D model and a picture automatically, and our team reviewed every result and asked for fixes.
- 2. Chat-driven design.** Later we used Zoo Design Studio and its AI agent, Zookeeper, driven almost entirely by typing instructions in plain language rather than by drawing.

What we learned

Part by part beats all at once.

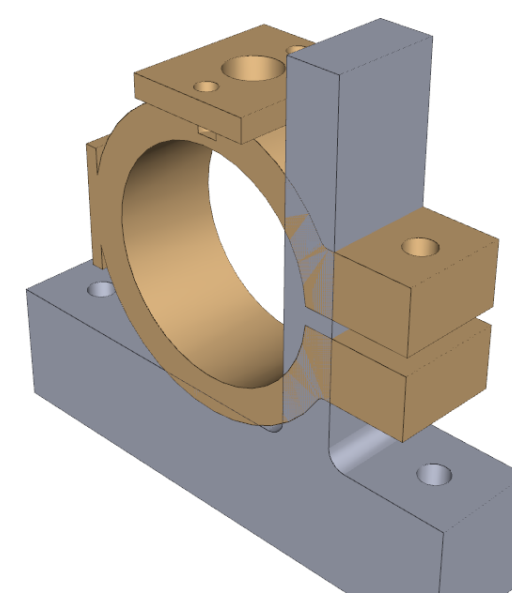
Asking the AI for the whole machine gave parts that overlapped or floated with no room for motors. Asking for *one part at a time*, with clear instructions on how it connects to its neighbours, worked.

The AI needs constant checking.

It would quietly change parts we had already finished, so every new version had to be re-inspected. Automated checks plus human review caught the problems.

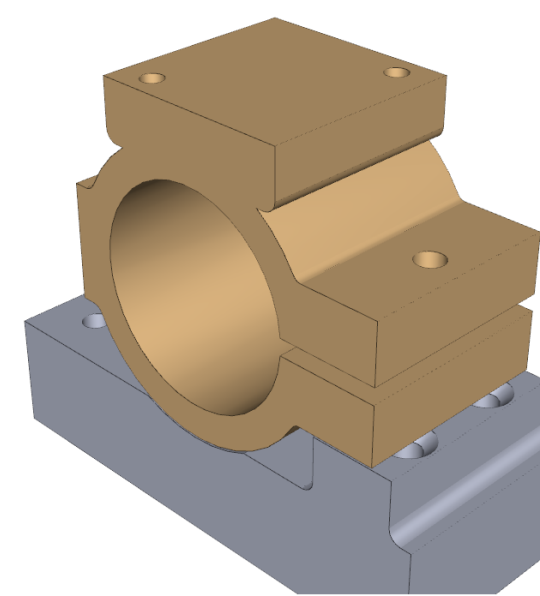
Honest about the trade-off.

An expert in normal CAD would have been faster and cleaner. What AI bought instead was a complete written record of every decision, prompt, and mistake—making the whole project reproducible.



First AI attempt:

impossible to build



After review:

a part that prints and works

Conclusion & what’s next

We designed and built a real laboratory instrument **entirely with generative AI, under human direction, and without ever opening conventional CAD software**. Because the full project history—every prompt, every generated part, and every failure—is public, the build doubles as a case study of what AI can and cannot yet do in hardware design. We have early CAD for a multi-powder version and continue to test new AI design tools.

Everything is open

Design files, firmware, and the complete log of AI prompts and responses are freely available:

[github.com/vertical-cloud-lab/
powder-doser](https://github.com/vertical-cloud-lab/powder-doser)

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